

Boundary-Aware Agricultural Field Segmentation

Does auxiliary distance prediction help more than direct boundary weighting?

HLCV Team 004

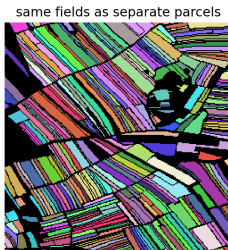
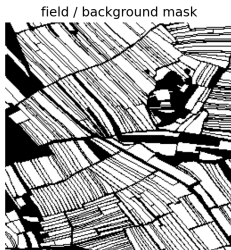
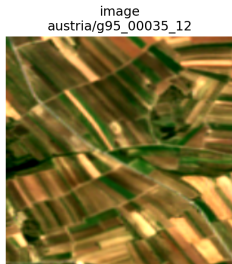
Saarland University

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Three controlled methods

1,430 held-out test chips

The challenge is preserving individual fields, not only field area



Failure mode

A binary mask can cover the correct area while merging neighboring fields.

Project goal

Improve border agreement without sacrificing region overlap.

Source: FTW imagery and labels; visualization generated in this project.

Data and supervision

Fields of The World subset

Countries	Austria, Croatia, Denmark
Input	two seasonal images, each with RGB+NIR (8 channels total)
Resolution	256 × 256 pixels; 10 m per pixel
Labels	field mask + FTW instance mask

10,938 train

1,347 validation

1,430 test

How the labels are used

Binary field mask

Mask head

Direct target: field or background at every pixel.

FTW instance mask

Provided by the dataset; separates neighboring fields.

converted into ↓

Signed distance field (SDF)

Distance head

Derived target: distance to the nearest field boundary.

Source: Kerner et al., "Fields of The World," AAAI 2025.

Related work and what we contributed

Built upon

- ▶ **FTW**: global, multi-date Sentinel-2 field benchmark.
- ▶ **U-Net**: encoder–decoder with skip connections for dense prediction.
- ▶ **Distance-aware supervision**: geometry can complement region losses.

Our project work

- ▶ Added an instance-derived SDF target and a second regression head.
- ▶ Added an SDF-based weighting map to focus BCE near field borders.
- ▶ Implemented matched baselines, Boundary IoU, checkpoint comparison, and test visualization.

Research question: where should boundary information enter the learning objective?

Source: Ronneberger et al., MICCAI 2015; Kervadec et al., MIDL 2019 / MedIA 2021.

Three methods, one controlled comparison

Method	Architecture	Training loss	Boundary signal
Mask baseline	one mask head	$\mathcal{L}_{BCE} + \mathcal{L}_{Dice}$	none
Two-head SDF	mask + distance heads	$\mathcal{L}_{BCE} + \mathcal{L}_{Dice} + 0.1\mathcal{L}_{SDF}$	auxiliary geometry
Boundary BCE	one mask head	$\mathcal{L}_{WBCE} + \mathcal{L}_{Dice}$	direct mask gradients

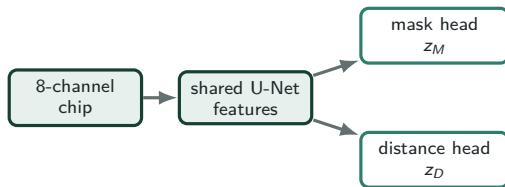
Held constant

Data, augmentation, U-Net width, optimizer, and evaluation code.

Checkpoint audit

The trained mask baseline uses ordinary **BCE + Dice**; it is not a Dice-only checkpoint.

Method 1: a second head predicts signed distance



Hypothesis

The SDF task forces shared features to encode individual field geometry.

- ▶ $D > 0$: inside an individual field
- ▶ $D \approx 0$: field boundary
- ▶ $D < 0$: background
- ▶ Inference still uses the mask head

+9,313 parameters

+0.12%

Joint objective

$$\mathcal{L}_{2H} = \mathcal{L}_{BCE}(z_M, Y) + \mathcal{L}_{Dice}(\sigma(z_M), Y) \\ + 0.1 \text{SmoothL1}(\tanh(z_D), D)$$

Method 2: put boundary emphasis directly into BCE

Boundary-weighted BCE

$$\mathcal{L}_{WBCE} = \frac{\sum_i w_i \text{BCE}(z_i, y_i)}{\sum_i w_i}$$
$$w_i = 1 + 20 \exp\left(-\frac{|D_i|}{0.12}\right)$$

$$\mathcal{L}_{final} = \mathcal{L}_{WBCE} + \mathcal{L}_{Dice}$$

Dice preserves global region overlap; weighted BCE changes which pixel errors dominate the gradient.

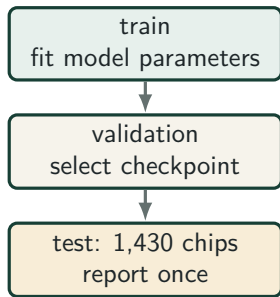
Pixel weight by distance

$ D $	BCE weight
0	21.0
0.12	8.36
0.24	3.71
far from border	≈ 1.0

Key difference

The boundary signal acts on the **final mask output**, not through an auxiliary task.

Evaluation protocol: model selection and test stay separate



Fair comparison

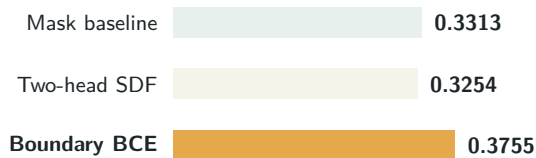
- ▶ Common probability threshold: **0.60**
- ▶ Boundary band radius: **2 pixels**
- ▶ Same 1,430 test chips for all models

Metrics

mIoU: region overlap

Boundary IoU: overlap of predicted and target boundary bands

Test result: direct boundary weighting is the clear winner



Boundary IoU @ threshold 0.60

Method	mIoU	BloU	Δ BloU
Baseline	.6572	.3313	–
2-head SDF	.6500	.3254	–0.59 pp
Boundary BCE	.6675	.3755	+4.42 pp

Main test finding

Boundary-weighted BCE improves both the target boundary metric and overall region overlap.

What did the second head actually contribute?

SDF alone

Earlier validation runs suggested a small gain, but the selected SDF checkpoint did **not** improve the held-out test baseline.

Test @ 0.60	mIoU	BloU
Baseline	.6572	.3313
Two-head SDF	.6500	.3254

Boundary BCE with vs. without SDF head

Test @ 0.60	mIoU	BloU
With SDF head	.6676	.3739
Without SDF head	.6675	.3755

Once the mask loss directly emphasizes borders, the auxiliary regression head adds parameters but no measurable test benefit.

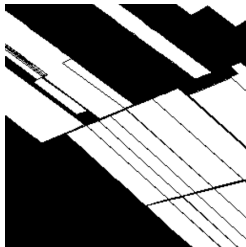
Interpretation: useful boundary information must reach the segmentation gradients directly.

Qualitative validation example: Austria

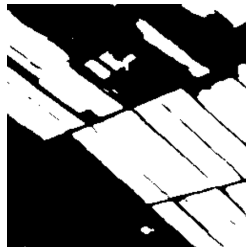
Sentinel-2 RGB



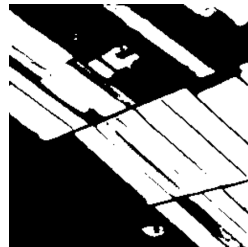
Ground-truth mask



Baseline
segmentation



Best-model
segmentation



The best model preserves more of the thin separations between neighboring fields.

Source: FTW validation chip `austria/g83_00040_14`; predictions generated in this project.

Conclusion and outlook

Takeaway

For this dataset and training setup,
boundary-weighted BCE + Dice is more effective
than adding an SDF regression head.

- ▶ Boundary IoU: 0.3313 $\rightarrow$ 0.3755 (+4.42 pp)
- ▶ mIoU: 0.6572 $\rightarrow$ 0.6675 (+1.03 pp)
- ▶ The simpler, mask-only boundary-BCE model is the final choice.

Directly optimize the error that matters at test time.

Next steps

- ▶ repeat with multiple random seeds
- ▶ report per-country performance
- ▶ test instance-separation post-processing
- ▶ study learned boundary calibration

Backup: threshold sensitivity on the test set

Method	0.40		0.50		0.60		0.70	
	mIoU	BloU	mIoU	BloU	mIoU	BloU	mIoU	BloU
Baseline	.6593	.3049	.6587	.3199	.6572	.3313	.6506	.3359
2-head SDF	.6554	.3017	.6528	.3153	.6500	.3254	.6369	.3246
Boundary BCE	.6706	.3364	.6731	.3618	.6675	.3755	.6520	.3682

Why the main talk uses 0.60

It is a single common operating point for all three models. The conclusion is unchanged across the tested thresholds.

Backup: full boundary-BCE objective and SDF ablation

General dual-head objective

$$\mathcal{L} = \lambda_B \mathcal{L}_{WBCE} + \lambda_D \mathcal{L}_{Dice} + \lambda_S \mathcal{L}_{SDF} + \lambda_{TV} \mathcal{L}_{TV}$$

Best simple model

$$\lambda_B = 1, \lambda_D = 1, \lambda_S = 0, \lambda_{TV} = 0$$

$$\alpha = 20, \sigma = 0.12$$

Full two-head variant

$$\lambda_B = 1, \lambda_D = 1, \lambda_S = 0.1, \lambda_{TV} = 0$$

Test BloU: 0.3739 vs. 0.3755 without SDF

Normalizing weighted BCE by $\sum_i w_i$ keeps its scale comparable while changing the relative importance of boundary pixels.

Backup: references

-  H. Kerner et al.
Fields of The World: A Machine Learning Benchmark Dataset for Global Agricultural Field Boundary Segmentation.
AAAI, 2025.
-  O. Ronneberger, P. Fischer, and T. Brox.
U-Net: Convolutional Networks for Biomedical Image Segmentation.
MICCAI, 2015.
-  H. Kervadec et al.
Boundary Loss for Highly Unbalanced Segmentation.
MIDL, 2019; extended in Medical Image Analysis, 2021.

All prediction panels and project diagrams were generated from this repository; no external figures are reproduced.